Satish Jhanwer

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Professional Summary

Senior Frontend Architect and **Engineering Lead** with **16+ years** of experience building scalable, accessible web platforms, micro-frontend frameworks, and multi-tenant design systems across **fintech, banking, and healthcare domains**. Architected a **150+ component React library** adopted across enterprise loan origination portals, and the MCA (Multi-Channel Application) **UI Framework powering 400+ channel** applications spanning web, mobile, ATM, POS, and back-office at Barclays. **Delivered AI/ML solutions including RAG pipelines, NLP classifiers, and predictive ML models at United Health Group. Author of XAA (Explainable AI for Accessibility Auditing)**, an ML-powered WCAG 2.1 auditing system combining a modular rule engine, BLIP-based image captioning, and automated fix suggestions. **M.Tech** in Artificial Intelligence and Machine Learning from **IIT Jodhpur**. Deep expertise in **React**, **TypeScript**, **Node.js**, **monorepo tooling**, **WCAG 2.1** AA accessibility, and **cross-functional team enablement**.

Experience

Senior Software Engineering Lead | United Health Group | Bangalore, India **Sep 2025 - Present**

- Engineered an IAM and OAuth 2.0 custom UI solution handling authentication and authorization flows, built on React.js within an Nx monorepo with a custom internal component library.
- Led development of the core UI platform, establishing shared infrastructure, component standards, and CI/CD workflows across product squads.
- Partnered with Product and Design teams in agile sprints to prototype and iterate UI solutions, **reducing design-to-dev handoff time by 25%**.
- Engineered WCAG 2.1 AA accessibility compliance into all components, performing automated audits using Axe and Lighthouse and remediating violations.
- Provided technical guidance to junior developers on Node.js backend best practices, code reviews, and Express.js middleware patterns.

Associate Architect | Sigma Infosolutions | Jodhpur, India **Jan 2022 - Sep 2025**

- Led a platform team of 15 engineers as Frontend Architect, owning end-to-end design and delivery of shared UI infrastructure for a large-scale Loan Origination System (LOS).
- Architected and delivered KCL (Kendra Component Library), a React.js and Material UI based design system of 150+ reusable, configurable, multi-tenant components supporting RBAC, SSO, and theming across all LOS portals.
- Built and owned the Borrower Portal, Backoffice Portal, and Merchant Portal configuration-driven, multi-tenant React.js applications with RBAC and SSO integration, backed by a Golang microservices architecture (DGraph, RabbitMQ, Docker, CDC pattern).
- Integrated CI/CD pipelines (GitHub Actions, Jenkins) for automated releases and semantic versioning of KCL, eliminating manual release overhead across 10+ consuming application teams.
- Authored TypeScript definitions, comprehensive documentation, and example applications, **reducing squad onboarding time by 40%**.
- Implemented automated testing with Jest and React Testing Library, achieving **95% component coverage** and preventing production regressions.

Senior UI Developer | Barclays LLC | Pune, India **Dec 2018 - Jan 2022**

- Architected and delivered the MCA (Multi-Channel Application) UI Framework a TypeScript and Node.js based framework enabling any client channel (web, mobile, ATM, POS, back-office) to integrate any frontend technology (React, Vue, Angular) as independently deployable, channel-agnostic modules.
- Scaled the MCA framework to power 400+ channel applications across Barclays's enterprise banking platforms, serving web browsers, mobile apps, ATMs, POS terminals, and back-office systems.

- Developed enterprise banking UIs for client onboarding, KYC compliance, loan origination, and trade reconciliation platforms used by 1000+ internal users.
- Modernized legacy dashboards using React, Redux, and React Query, achieving **40% improvement in load times** and **25% higher workflow completion rates**.
- Built reusable components for audit trail visibility, approval workflows, and multi-step form validation critical for regulatory submissions.
- Delivered WCAG 2.1 AA accessibility across all channel applications via systematic Axe and Lighthouse auditing, establishing accessibility baselines for the MCA platform.
- Designed a scalable UI framework **improving user satisfaction by 30%** and **reduced cross-team issue resolution time by 40%**.

Senior Software Engineer | Sigma Infosolutions | Ahmedabad, India **Apr 2013 - Dec 2018**
- Developed and deployed user interfaces, enhancing web application functionality and **increasing user engagement by 25%**.
- Conducted technical sessions that helped junior developers **increase debugging efficiency by 30%**.
- Championed adoption of modern frontend technologies through market analysis, **cutting project costs by 15%**.
- Coordinated team efforts to deliver high-priority projects, achieving 100% on-time completion rate.

Software Engineer | Sigma Infosolutions | Bangalore, India **Apr 2012 - Mar 2013**
- Developed, tested, and debugged full-stack applications using JavaScript, jQuery, Groovy, HTML, and CSS, **improving performance by 20%**.
- Proposed and implemented improvements aligned with industry standards, resulting in **15% increase in operational efficiency**.

Associate Software Engineer | Sigma Infosolutions | Bangalore, India **Oct 2010 - Mar 2012**
- Developed, tested, and debugged full-stack applications using JavaScript, jQuery, Groovy, HTML, and CSS.

Skills

Languages | TypeScript, JavaScript (ES6+), Python, HTML5, CSS3, Groovy
Frameworks | React.js, Vue.js, Node.js, Express.js, GraphQL
AI/ML | Explainable AI (XAI), RAG Pipelines, LangChain, Vector Databases, Classical NLP, Scikit-learn, BLIP (Image Captioning), Predictive Modeling, Flesch-Kincaid Readability, Machine Learning Workflows
Component and State | Redux, Context API, React Query, React Hook Form, React Router, Hooks, Functional Components, Material UI
UI Tooling | Storybook, Ag-Grid React, Webpack, Babel, Vite
Monorepo and Build | Nx, Lerna, Turborepo, npm, Yarn
Micro-Frontend Frameworks | Module Federation, MCA Framework, Single-SPA patterns, Channel-Agnostic Architecture
Testing | Jest, React Testing Library, Cypress, Mocha, Jasmine, Chai, Supertest
Accessibility | WCAG 2.1 AA, ARIA, Axe, Lighthouse, Automated Accessibility Auditing
Backend and APIs | REST APIs, Socket.io, GraphQL, Node.js Streams, PM2, JWT
Cloud and DevOps | Docker, Kubernetes, AWS, Nginx, CircleCI, GitHub Actions, Jenkins, GitLab CI
Auth and Security | IAM, OAuth 2.0, RBAC, SSO
Methodologies | Agile/Scrum, TDD (Test-Driven Development), BDD, CDC Pattern

Education

M.Tech in Artificial Intelligence & Machine Learning

Indian Institute of Technology(IIT), Jodhpur, India

2024-2026 | CGPA: 9.09 Ranked 1st

B.Tech in Information Technology

Rajasthan Technical University, Kota, India

2006-2010 | 75% with honours

Projects

XAA - Explainable AI for Accessibility Auditing | M.Tech Thesis Project | IIT Jodhpur
- Designed and built an end-to-end explainable accessibility auditing system for web content, going beyond Axe and Lighthouse by pairing WCAG 2.1 compliance checks with human-readable explanations, WCAG-linked references, and actionable fix suggestions.
- Engineered a modular rule engine using a decorator-based RuleRegistry pattern for extensible WCAG 2.1 compliance evaluation across parsed HTML structures.
- Integrated BLIP model for automated alt-text generation on images and Flesch-Kincaid scoring for readability analysis, enabling ML-powered accessibility insights.
- Delivered ML-powered color contrast fix suggestions with interactive Fix It previews and exported reports in JSON, CSV, PDF, and HTML formats.
- Built a React dashboard with dark mode, project management, audit history, trend tracking, and benchmarking against axe-core, secured via JWT authentication and RBAC.
- Implemented a 7-layer CI quality assurance pipeline via GitHub Actions, covering linting, unit tests, integration tests, accessibility checks, build validation, security scans, and report generation.

Retrieval-Augmented Generation (RAG) pipeline | United Health Group | Healthcare
- Architected a Retrieval-Augmented Generation (RAG) pipeline for clinical intelligence, enabling semantic search and context-aware responses over structured and unstructured medical data using vector databases and LLM integrations.

Mental Health Concern Classifier | United Health Group | Healthcare
- Built a classical NLP workflow to classify mental health concerns from text inputs, encompassing data preprocessing, feature engineering, and model evaluation pipelines.

Predictive Modeling for Treatment Cost Estimation | United Health Group | Healthcare
- Developed an end-to-end machine learning workflow using Python and Scikit-learn to predict patient treatment costs, including feature selection, model training, and validation.

LOS Platform (Loan Origination System) | Sigma Infosolutions | Fin-Tech
- Built the full frontend platform for an enterprise LOS: KCL component library (150+ components), Borrower Portal, Backoffice Portal, and Merchant Portal all configuration-driven, multi-tenant React.js applications backed by Golang microservices on Docker/Kubernetes/AWS.

MCA UI Framework | Barclays | Banking
- Architected a channel-agnostic Multi-Channel Application (MCA) framework in TypeScript and Node.js, enabling React, Vue, and Angular applications to be deployed as independent modules across web, mobile, ATM, POS, and back-office channels adopted by 400+ channel applications enterprise-wide.

Awards and Accomplishments

- **Influencer Leader of the Year Award**, 2023
- **Quarterly Appreciation Award**, 2016
- **Spot Award**, 2015
- **Best Debut Employee Award**, 2011